

Cardiac MRI audit presentation script

Ten-slide speaker script | 787 spoken words | Approximately 8-9 minutes with brief pauses

Slide 1: Auditing Deep Learning for Cardiac MRI Through Semantic Transitions and Conflict-Aware Production Rules

Today I will present our study of how semantic transitions and conflict-aware production rules can make cardiac magnetic resonance analysis more auditable. This work was conducted by Pavlo Radiuk, Oleksander Barmak, and Iurii Krak. We connect image-derived descriptors to recognizable cardiac measurements, then to rules that expose supported conclusions, conflicts, and abstention. Although the framework is intended for deep-learning auditing, the reported experiments evaluate deterministic descriptors rather than a trained deep encoder.

Slide 2: Prediction quality leaves an audit gap

Strong predictive performance does not, by itself, explain whether a cardiac measurement or rule is trustworthy. The published systems summarized here provide context for diagnostic accuracy and measurement error, while our study emphasizes traceable audit outputs. Their results come from different splits and objectives, so the tables should not be read as a controlled ranking. The practical question is how to connect recognizable measurements with reusable rules, evidence provenance, and an explicit option to abstain.

Slide 3: Make every audit output traceable

Our objective is to preserve a traceable path from each patient record through semantic reconstruction to rule-based auditing. We used 150 cases from the Automated Cardiac Diagnosis Challenge, or ACDC, split into 85 training, 15 validation, and 50 test cases. Another 100 cases from the Multi-Centre, Multi-Vendor and Multi-Disease cohort, or M&Ms, provided an untouched external evaluation. All 20 programmed semantic validation checks passed. However, 92 cases across the cohorts still awaited manual review; these are pending assessments, not confirmed errors or completed expert validation.

Slide 4: A frozen, patient-level audit chain

The architecture follows the patient from cine images and masks through formal descriptors, reconstructed cardiac measurements, and guarded rule outputs. The evaluated representation maps 57 deterministic descriptors to 128 coordinates. Although the original framework diagram names weighted entropy-density discretization, or WEDD, this release substitutes frozen training tertiles and does not validate the full WEDD method. The eight-step procedure preserves this sequence while adding quality control, predicate alignment, fidelity assessment, and perturbation testing.

Slide 5: Regularized semantic reconstruction

Ridge regression provides an inspectable mapping from the formal representation to 28 cardiac measurements, including volumes, ejection fractions, mass, and wall thickness. Preprocessing is learned exclusively from the training cases to prevent information leakage. Five-fold training cross-validation selected a regularization penalty of three, after which the transition and preprocessing were frozen. Training-derived tertiles then assign each eligible measurement to a low, middle, or high state. The transition coefficients describe statistical associations, not causal physiological effects.

Slide 6: Rules, guards, and clinical alignment

Rules combine at most three conditions and require support from at least three training cases, with confidence of at least sixty percent. A rule can produce a conclusion only when its evidence and validity requirements are satisfied. Severe quality problems, unstable reconstruction, missing prerequisites, unmatched cases, or contradictory conclusions trigger abstention with a recorded reason. Physician-style similarity scores help align eligible rules with clinical predicates, accounting for conflict, missing evidence, and instability. Crucially, a high similarity score never overrides the abstention guard.

Slide 7: Semantic fidelity and external domain shift

Internal reconstruction was strongest for selected targets: correlations reached 0.928 for end-diastolic wall thickness and 0.863 for left ventricular ejection fraction. However, normalized reconstruction error rose from 0.767 on the held-out ACDC test set to 1.726 externally, a 2.25-fold increase. External bias also reached minus 136 milliliters for left ventricular end-diastolic volume and minus 96.6 grams for myocardial mass. The target-level confidence intervals help contextualize internal precision. Together, these results show that strong internal associations do not establish reliable performance across domains.

Slide 8: Coverage must accompany rule accuracy

The comparator uses measured cardiac semantics, rather than rule outputs, and achieved 86 percent test accuracy, macro-F1 of 0.861, and an area under the receiver operating characteristic curve of 0.973. By contrast, the 200-rule audit covered 32 of 50 test cases and abstained on 18. Balanced accuracy was 0.644 and macro-F1 was 0.612 among non-abstained cases. Conflict affected 31.3 percent of covered cases, while the comparator's expected calibration error was 0.179. Selective accuracy must therefore be interpreted alongside coverage, conflict, and uncertainty.

Slide 9: Predicate alignment and perturbation sensitivity

Predicate alignment was strongest for the abnormal right ventricular proxy, with similarity of approximately 0.90. Valvular and congenital proxies remained review-only, while unsupported disease families were left unmatched. These scores indicate compatibility with specified predicates, not clinician agreement or diagnostic validation. Across 48 perturbation runs on 12 cases, a two-degree rotation caused the largest drift despite nearly preserved mask counts. Because these exploratory changes mix semantic units, they identify orientation sensitivity rather than a clinical tolerance, motivating orientation normalization before future evaluation.

Slide 10: Transparent auditing and next validation

This study establishes a traceable audit chain that exposes semantic reconstruction, external degradation, conflicting rules, and abstention. Its present role is selective research auditing, with substantial limitations from domain shift, rule sparsity, and incomplete expert review. The next stage should evaluate frozen deep encoders, full WEDD, orientation normalization, vendor recalibration, and prospective clinician use. These steps are needed before drawing conclusions about clinical value. Thank you for your attention; our authors' contact details are shown here, and we welcome questions.